

Differentiation

→ Differentiate the following w.r.t. x (Division Rule)

① $y = \frac{x^2-1}{x^2+1}$

④ $y = \frac{a+b\sin x}{a\sin x+b}$

② $y = \frac{1+\sin x}{1-\sin x}$

⑤ $y = \frac{a+b\cos x}{b+a\cos x}$

③ $y = \frac{1+\tan x}{1-\tan x}$

⑥ $y = \frac{\log x}{x}$ at $x=1$

→ Differentiate the following w.r.t. x (chain Rule)

① $y = \log(\sec x + \tan x)$

⑥ $y = \log x^3 \cdot \sin(\log x)$

② $y = \log\left(\frac{x^2+1}{x^2-1}\right)$

⑦ $y = e^{\sin x} \sec 2x$

③ $y = \log\left(\frac{\sin x}{1+\cos x}\right)$

⑧ $y = e^{3x} \cdot \cos 2x$

④ $y = \log\left(\sqrt{\frac{1+\sin x}{1-\sin x}}\right)$

⑨ $y = \frac{\sin(\log x)}{x}$

⑤ $y = \log(x + \sqrt{x^2+a^2})$

→ Differentiate the following w.r.t. x (Implicit function)

① $x^3 + y^3 = 3axy$

④ $y = \cos(x+y)$

② $x+y = \sin(xy)$

⑤ $y = \sin(x+y)$

③ $x \sin y + y \sin x = 5$

⑥ $x^2 + xy + y^2 = 10$

→ Differentiate the following w.r.t. x (Parametric)

① $x = a \sec \theta$ $y = b \tan \theta$

⑤ $x = a \cos^4 \theta$ $y = b \sin^4 \theta$

② $x = a \sin \theta$ $y = a(1+\cos \theta)$

⑥ $y = a(\theta + \sin \theta)$ $x = a(1+\cos \theta)$

③ $x = a(\theta + \sin \theta)$ $y = a(1-\cos \theta)$

⑦ $x = \frac{1}{2}\left(t + \frac{1}{t}\right)$ $y = \frac{1}{2}\left(t - \frac{1}{t}\right)$

④ $x = a(t + \cos t)$ $y = a(1 + \sin t)$

→ Differentiate the following w.r.t. x (Logarithmic)

① $y = x^x$

④ $y = (\sin x)^x + x^{\cos x}$

② $y = (\sin x)^x$

⑤ $y = x^{\sin x} + x^x$

③ $y = (\sin x)^{\tan x}$

⑥

Successive differentiation

① If $y = 2e^{3x} + 3e^{-2x}$ then prove that

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 0$$

② For $y = \cos Ax + \sin Ax$ show that

$$\frac{d^2y}{dx^2} + A^2y = 0$$

③ If $y = a \cos(\log x) + b \sin(\log x)$ show that

$$x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0$$

④ If $y = A \cos pt + B \sin pt$ then prove that

$$\frac{d^2y}{dt^2} + p^2y = 0$$

⑤ If $y = \log(\sin x)$ then prove that

$$\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + 1 = 0$$

⑥ If $y = \log(x + \sqrt{1+x^2})$, prove that

$$(1+x^2) \frac{d^2y}{dx^2} + x \frac{dy}{dx} = 0$$

⑦ If $x = \sin t$ and $y = \sin pt$ then prove that

$$(1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} + p^2y = 0$$

⑧ If $y = \sin(\sin x)$, prove that

$$y_2 + y_1 \tan x + y \cos^2 x = 0$$

⑨ If $y = e^{m \tan^{-1} x}$ then prove that

$$(1+x^2) \frac{d^2y}{dx^2} + (2x-m) \frac{dy}{dx} = 0$$

Application of Derivative

- ① The eqⁿ of motion of a particle is $s = t^3 - 3t^2 + 4t + 3$. Find its velocity and acceleration at $t = 2$
- ② Equation of motion of a particle is $s = t^3 - 6t^2 + 8t - 4$ then find the velocity and acceleration of the moving particle at $t = 3$ second
- ③ The equation of motion of a particle is $s = t^3 + 3t$, $t > 0$.
 - ① Find the velocity and acceleration at $t = 3$?
 - ② When do velocity and acceleration become equal?
- ④ The equation of motion of a particle is given by $s = t^3 - 6t^2 + 9t + 6$. Find the velocity of the particle when its acceleration is zero.
- ⑤ If the equation of motion of a particle is $s = t^3 - 6t^2 + 9t + 6$. Find its velocity when $t = 0$. Also find its acceleration when $v = 0$
- ⑥ The equation of motion of a particle is $s = t^3 - 6t^2 + 9t + 6$ where s is in meter and t is in second.
 - ① Find v and a when $t = 2$ sec
 - ② Find s when particle changes its direction.
- ⑦ Eqⁿ of motion of a particle is $s = t^3 - 5t^2 + 3t$. When particle come to rest? Find acceleration at that time?

Maxima & Minima

Find maximum and minimum values of the function

- ① $f(x) = x^3 - 3x + 11$
- ② $f(x) = 3x^3 - 4x^2 - x + 5$
- ③ $f(x) = 2x^3 - 15x^2 + 36x + 10$
- ④ $f(x) = 2x^3 - 3x^2 - 12x + 5$
- ⑤ $f(x) = x^3 - 4x^2 + 5x + 7$
- ⑥ $f(x) = x + \frac{7}{x}$
- ⑦ $f(x) = \sin x + \cos x$
where $x \in [0, 2\pi]$